CHAPTER-4

POLYMERIZATION

Q What do you mean by polymerization? Classify it.

Polymerization: Polymerization is a process through which a large number of monomer molecules react together to form a polymer. The macromolecules produced from a polymerization may have a linear or branched structure.

Classification of polymerization:

I. Condensation polymerization or step growth polymerization.
II. Addition or chain growth polymerization.
   a. Radical chain polymerization.
   b. Ionic polymerization
   c. Co-ordination polymerization.
III. Co-polymerization.

Q. What is meant by step-growth polymerization? write the characteristics of step growth polymerization [2020]

step-growth polymerization:

step-growth or condensation polymerization takes place between two bifunctional molecules to produce one larger polyfunctional molecule with the possible elimination of a small molecule such as H_2O, NH_3, HCl etc.

Example: polyester, polyamide, polycarbonates.

characteristics:

1. The features of step-growth polymerizations are that the mechanism is step by step chemical reaction with no dependence on previous steps.
2. No initiator needed.
3. Bifunctional reaction functional groups on the end.
4. Two active sites per monomer
5. Two different types of monomer reacting.

Q. What is Carother's equation? [2020] [2016]
or, Deduce Canother's equation. [2017]

Carothers equation:

In step growth polymerization, the Carothers equation gives the degree of polymerization, X_n, for a given fractional monomer conversion, p.

Consider a flask of monomer. If there are N_o no. of initially (N_o) reacted molecules in the flask and N, number of molecules remaining at time t. Then, the average degree of polymerization, $\bar{X}_n$ at time t is given by-

$$\bar{X}_n = \frac{N_o}{N} \quad \text{——— (1)}$$

Since the amount reacted is $N_o - N$ and we can define p as the "conversion" and fraction reacted then -

The extent of reaction -

$$p = \frac{\text{No. of group which have reacted}}{\text{No. of group initially present}}$$

$$\Rightarrow p = \frac{N_o - N}{N_o}.$$

$$\Rightarrow p = 1 - \frac{N}{N_0}$$

$$\Rightarrow \frac{N}{N_0} = 1 - p$$

$$\Rightarrow \frac{N_0}{N} = \frac{1}{1-p} \quad \text{---(11)}$$

From equation (1) and (11) -

The average degree of polymerization, $\bar{X}_n = \frac{1}{1-p}$

This is known as Carothers equation.

Q. Write the mechanism of cationic polymerization. [2017] [2018] [2020]

Let, A is a catalyst, RH is a co-catalyst and M is the monomer.

Initiation :

$$A + RH \rightleftharpoons H^+ A\bar{R} \quad \text{ion pair formation}$$

$$M + \overset{+}{H} M \bar{A}R \xrightarrow{k_i} H\overset{+}{M}M\bar{A}R$$

propagation:

$$H\overset{+}{M}_x\bar{A}R + M \xrightarrow{k_p} H\overset{\oplus}{M}_{x+1}\overset{\ominus}{A}R$$

Termination:

$$H\overset{\oplus}{M}_x\overset{\ominus}{A}R \xrightarrow{k_t} M_x + H\overset{\oplus}{A}\overset{\ominus}{R}$$

$$H\overset{\oplus}{M}_x\overset{\ominus}{A}R + M \xrightarrow{k_{tr}} M_x + H\overset{\oplus}{M}\overset{\ominus}{A}R \quad \text{(chain transfer to monomer)}$$

The rate of initiation,

$$v_i = k k_i [A][RH][M]$$

The rate of termination,

$$v_t = k_t [H\overset{\oplus}{M}\overset{\ominus}{A}R]$$

$$= k_t [\overset{\oplus}{M}]$$

Applying steady state approximation,

$$v_i = v_t$$

$$\Rightarrow k k_i [A][RH][M] = k_t [\overset{\oplus}{M}]$$

$$\Rightarrow [\overset{\oplus}{M}] = \frac{k k_i}{k_t} [A][RH][M]$$

Rate of propagation,

$$v_p = k_p [M][\overset{\oplus}{M}]$$

$$= k_p [M] \frac{k k_i}{k_t} [A][RH][M]$$

$$= k \frac{k_i k_p}{k_t} [A][RH][M]^2$$

Kinetic chain length,

$$\bar{x}_n = \frac{\text{Rate of propagation}}{\text{Rate of formation of dead chain}}$$

$$= \frac{k_p [\dot{M}] [M]}{k_t [\dot{M}] + k_{tr} [\dot{M}] [M]}$$

If termination predominate over transfer,

$$\bar{x}_n = \frac{k_p [\dot{M}] [M]}{k_t [\dot{M}]}$$

$$= \frac{k_p}{k_t} [M]$$

Whereas, if transfer predominate,

$$\bar{x}_n = \frac{k_p [\dot{M}] [M]}{k_{tr} [\dot{M}] [M]}$$

$$= \frac{k_p}{k_{tr}}$$

Applying Arrhenius theory,

$$k_i = A_i e^{-E_i/RT}$$

$$k_p = A_p e^{-E_p/RT}$$

$$k_t = A_t e^{-E_t/RT}$$

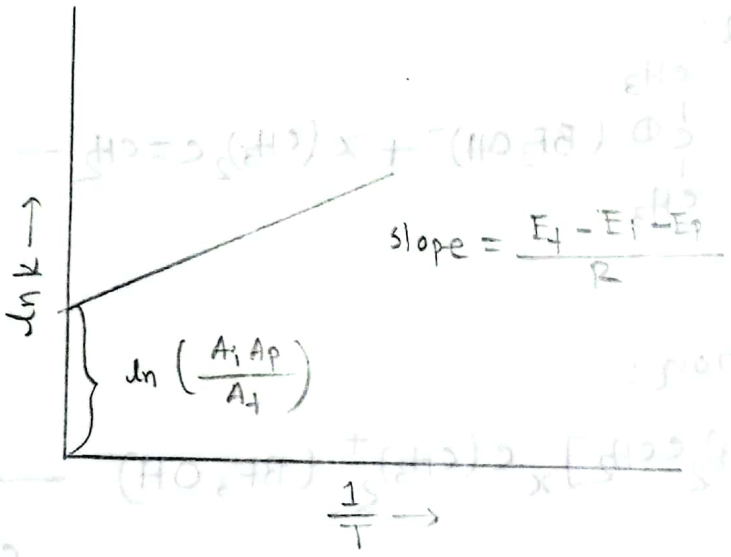
Overall rate constant,

$$k_R = \frac{A_i A_p}{A_t} \cdot \frac{e^{-E_i/RT} \cdot e^{-E_p/RT}}{e^{-E_t/RT}}$$

$$\Rightarrow k_R = \frac{k_i k_p}{k_t} = \frac{A_i A_p}{A_t} e^{(E_t - E_i - E_p)/RT}$$

$$\Rightarrow \ln k_R = \ln\left(\frac{A_i A_p}{A_t}\right) + \frac{E_t - E_i - E_p}{RT}$$

$$\Rightarrow \ln k_R = \frac{E_t - E_i - E_p}{RT} + \ln\left(\frac{A_i A_p}{A_t}\right)$$



The overall rate of polymerization increases with decreasing temperature. If $E_t > E_i + E_p$ and the molecular weight increases as the temperature of reaction decrease if $E_t > E_p$.

Mechanism (example):

Initiation:

1st step:

$$BF_3 + H_2O\ (\text{trace}) \rightleftharpoons H^+(BF_3OH)^-$$

catalyst — co-catalyst — catalyst + co-catalyst complex.

2nd step:

$$H^+(BF_3OH)^- + (CH_3)_2C{=}CH_2 \longrightarrow (CH_3)_3\overset{\oplus}{C}\,.\,(BF_3OH)^-$$

propagation:

$$CH_3{-}\overset{CH_3}{\underset{CH_3}{\overset{|}{\underset{|}{C}}}}{}^{\oplus}\ (BF_3OH)^- + x\,(CH_3)_2C{=}CH_2 \longrightarrow H_3C{-}\overset{CH_3}{\underset{CH_3}{\overset{|}{\underset{|}{C}}}}{-}CH_2{-}\overset{CH_3}{\underset{CH_3}{\overset{|}{\underset{|}{C}}}}{}^{\oplus}\ (BF_3OH)^-$$

Termination:

$$CH_3[(CH_3)_2CCH_2]_xC(CH_3)_2^+\ (BF_3OH)^- \longrightarrow$$

$$CH_3[(CH_3)_2CCH_2]_xC(=CH_2)(CH_3) + H^+(BF_3OH)^-$$

* Anionic polymerization:

Kinetics:

A generalised scheme for the anionic polymerization may be written as –

$$IA \xrightarrow{k_d} I^{+} \bar{A} \quad \text{(ion-pair formation)}.$$

Initiation:

$$A^{\ominus}I^{\oplus} + M \xrightarrow{k_i} A\overset{\ominus}{M}I^{\oplus}$$

propagation:

$$AM^{\ominus}I^{\oplus} + M \xrightarrow{k_p} AM\overset{\ominus}{M}I^{\oplus}$$

Termination:

$$AM_nM^{\ominus}I^{\oplus} + HA \xrightarrow{k_t} AM_nMH + A^{\ominus}I^{\oplus}$$

Where IA is the initiator and HA is a protonating agent.

Rate of initiation:

$$v_i = -\frac{d[M_i]}{dt} = k_i[IA][M]$$

Rate of termination:

$$v_t = -\frac{d[M_i]}{dt} = k_t[\overset{\ominus}{M}][HA]$$

Applying steady state approximation -

$$k_t [M^{\oplus}] [HA] = k_i [IA] [M]$$

$$\Rightarrow [M^{\oplus}] = \frac{k_i}{k_t} \frac{[IA] [M]}{[HA]}$$

Rate of propagation,

$$v_p = k_p [M] [M^{\oplus}]$$

$$= k_p [M] \times \frac{k_i}{k_t} \frac{[IA] [M]}{[HA]}$$

$$= \frac{k_i k_p}{k_t} \frac{[IA] [M]^2}{[HA]}$$

Degree of polymerization,

$$\bar{x}_n = \frac{\text{Rate of polymerization}}{\text{Rate of termination}}$$

$$= \frac{k_p [M] [M^{\oplus}]}{k_t [M^{\oplus}] [HA]}$$

$$= \frac{k_p}{k_t} \frac{[M]}{[HA]}.$$

9. What is the difference between cationic and anionic polymerization? [2018]

Cationic polymerization	Anionic polymerization.
1. Monomers have double bonds and the polymerization proceeds by an attack on the π-electron of the double bond by a cation (usually a carbonium ion).	1. Monomers have double bonds and the polymerization proceeds by an attack on the π-electron of the double bond by an anion. (usually a carbanion).
11. Living polymer is not possible here.	11. Living polymer is possible here.
111. polymer has a ~~too~~ medium molecular weight distribution. $\frac{\bar{M}w}{\bar{M}n}$ = 1.5 to 2	111. polymer has a very low molecular weight distribution $\frac{\bar{M}w}{\bar{M}n}$ is less than 1.5
iv. In cationic polymerization, Lewis acids and Friedel-craft catalyst such as $AlCl_3$, BF_3, $SnCl_4$, H_2SO_4 are used.	iv. Highly electropositive metals and their compounds are used initiator e.g. Li, BuLi, n-BuMgBr
v. The kinetic rate expression is – $V_p = k'[A][RH][M]^2$ where, A = catalyst, RH = co-catalyst, M = Monomer	v. The kinetic rate expression is – $V_p = k' \frac{[IA][M]^2}{[HA]}$ where, IA = initiator, M = monomer, HA = propagating agent solvent.

Q. What are living polymers? Explain with suitable example. [2018] [2016]

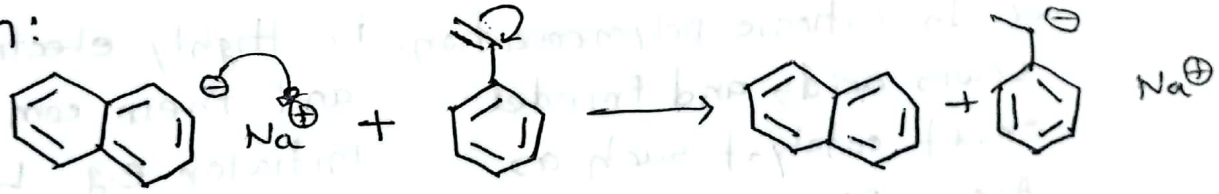
Living polymer:

Living polymers are those which have no termination step and no chain transfer. polymerization stop when there is no more monomer.

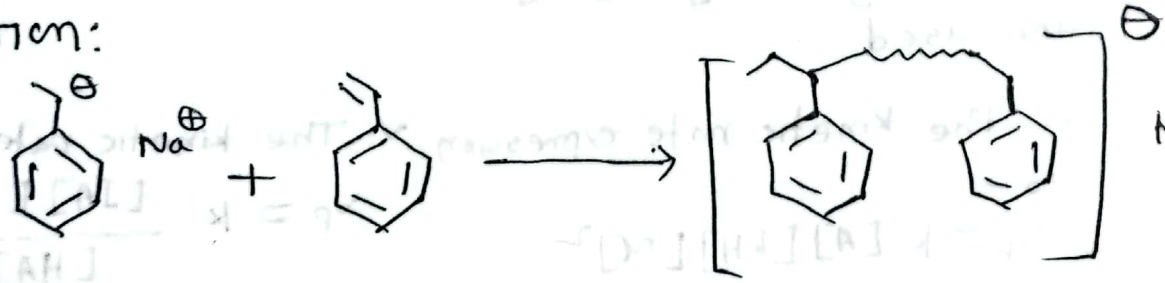
Anionic polymerization with purified reagents may lead to system in which termination is lacking. This type of anionic polymer are called living polymer.

Example of living polymer is anionic polymerization of sodium dihydronapthylide with styrene.

Initiation:



propagation:



Q. Derive rate expression for step Growth (condensation) polymerization in presence of acid catalyst. Show graphically which process is more efficient. [2017] [2016]

Kinetics of step growth polymerization in presence of acid catalyst:

$$\text{Acid} + \text{alcohol} \xrightarrow[\text{of acid catalyst}]{\text{In presence}} \text{polyester}$$

The rate of this reaction –

$$-\frac{d[COOH]}{dt} = k[\text{catalyst}][COOH][OH] = -\frac{d}{dt}[OH] \quad \text{——(I)}$$

When, $[OH] = [COOH] = c$

Equation (I) becomes –

$$-\frac{dc}{dt} = k[\text{catalyst}]c^2 \quad \text{——(II)}$$

Since catalyst is a constant, then equation (II) becomes –

$$-\frac{dc}{dt} = k_x c^2$$

$$\Rightarrow -\frac{dc}{c^2} = k_x dt \quad \text{——(III)}$$

Integrating equation (iii) using limit $0 \to t$ and $C_0 \to C$

$$-\int_{C_0}^{C} \frac{1}{C^2}\, dC = k_x \int_0^t dt$$

$$\Rightarrow -\left[\frac{C^{-1}}{-1}\right]_{C_0}^{C} = k_x t$$

$$\Rightarrow \left(\frac{1}{C} - \frac{1}{C_0}\right) = k_x t$$

$$\Rightarrow k_x t = \frac{1}{C} - \frac{1}{C_0} \quad \text{——(iv)}$$

putting $C = C_0(1-p)$ in equation (iv)

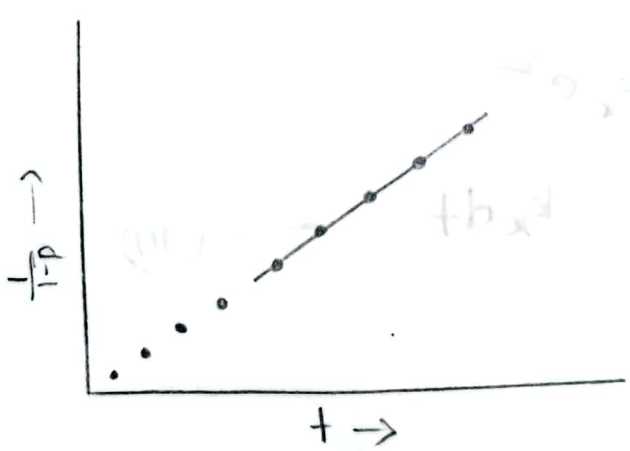
$$k_x t = \frac{1}{C_0(1-p)} - \frac{1}{C_0}$$

$$\Rightarrow k_x t = \frac{1}{C_0}\left\{\frac{1}{(1-p)} - 1\right\}$$

$$\Rightarrow k_x C_0 t = \frac{1}{(1-p)} - 1$$

$$\therefore \frac{1}{(1-p)} = k_x C_0 t + 1$$

Thus a plot of $\frac{1}{1-p}$ vs t given a straight line.



Degree of polymerization,

$$Dp = \frac{c_0}{c} = \frac{1}{1-p}$$

$$\Rightarrow Dp = k_x c_0 t + 1 \quad \text{——— (v)}$$

From equation (v), Dp appears proportional with time. This mean that the MW built up in this case will be faster.

In absence of catalyst:

Dibasic acid + alcohol $\xrightarrow[\text{acid catalyst}]{\text{In absence of}}$ polyester

The rate of the reaction –

$$-\frac{d[COOH]}{dt} = -\frac{d[OH]}{dt}$$

$$= k[COOH]^2[OH] \quad \text{——— (vi)}$$

If the concentration of carboxyl and hydroxyl group are same, i.e $[OH] = [COOH] = c$, then equation (vi) becomes –

$$-\frac{dc}{dt} = kc^3 \quad \text{——— (vii)}$$

$$\Rightarrow -\frac{dc}{dt c^3} = dt\,k \quad \text{——— (viii)}$$

Integrating equation (viii) using limit $0 \to t$ and $c_0 \to c$

$$-\int_{c_0}^{c} \frac{1}{c^3}\, dc = k\int_0^t dt$$

$$\Rightarrow -\left[\frac{c^{-2}}{-2}\right]_{c_0}^{c} = kt$$

$$\Rightarrow \frac{1}{2}\left(\frac{1}{c^2} - \frac{1}{c_0^2}\right) = kt$$

$$\Rightarrow 2kt = \frac{1}{c^2} - \frac{1}{c_0^2} \quad \text{——— (ix)}$$

Let, p is defined as fraction of functional group which undergoes esterification at time t. and is given ~~by~~ as –

$$p = \frac{c_0 - c}{c_0}$$

$$\Rightarrow p = 1 - \frac{c}{c_0}$$

$$\Rightarrow \frac{c}{c_0} = 1 - p$$

$$\Rightarrow c = c_0(1-p) \quad \text{——— (x)}$$

Putting the value of c in equation (ix)

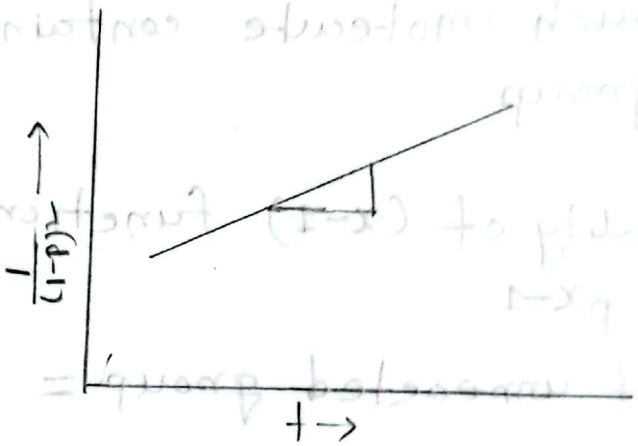
$$2kt = \frac{1}{c_0^2(1-p)^2} - \frac{1}{c_0^2}$$

$$\Rightarrow 2kt = \frac{1}{c_0^2}\left\{\frac{1}{(1-p)^2} - 1\right\}$$

$$\Rightarrow 2c_0^2 kt = \frac{1}{(1-p)^2} - 1$$

$$\Rightarrow \frac{1}{(1-p)^2} = 1 + 2c_0^2 kt \quad \text{——— (xi)}$$

Hence the plot of $\frac{1}{(1-p)^2}$ vs time, t gives a straight line.



Again, degree of polymerization

$$D_p = \frac{c_0}{c} = \frac{1}{(1-p)}$$

$$\Rightarrow D_p^2 = \frac{1}{(1-p)^2} = 2k c_0^2 t + 1 \quad \text{——— (xii)}$$

From (xii), D_p appears as the second power and time is first power and hence the polymer molecular weight increase only slowly.

Hence first pathway (presence of catalyst) is more efficient.

Q. Deduce number or mole fraction distribution for step Growth polymerization. [2017]

Let p is the probability that a functional group reacted at a time t. Thus the finding of functional group arrangement is (1−p).

Now, let a given molecule contains x repeating units and such molecule contains (x−1) reacted functional group.

Thus,

probability of (x−1) functional group is $= p^{x-1}$

probability of unreacted group $= 1-p$

Therefore, total probability for complete molecule $= p^{x-1}(1-p)$

If, there are N no. of molecules, total no. of x is given by −

$$N_x = N p^{x-1}(1-p) \quad \text{——— (1)}$$

If the total no. of unit present is N_0, then.

$$N = N_0(1-p)$$

$$\Rightarrow N_x = N_0(1-p)^2 p^{x-1} \quad \text{---(11)}$$

This is the number distribution function for a linear stepwise polymerization.

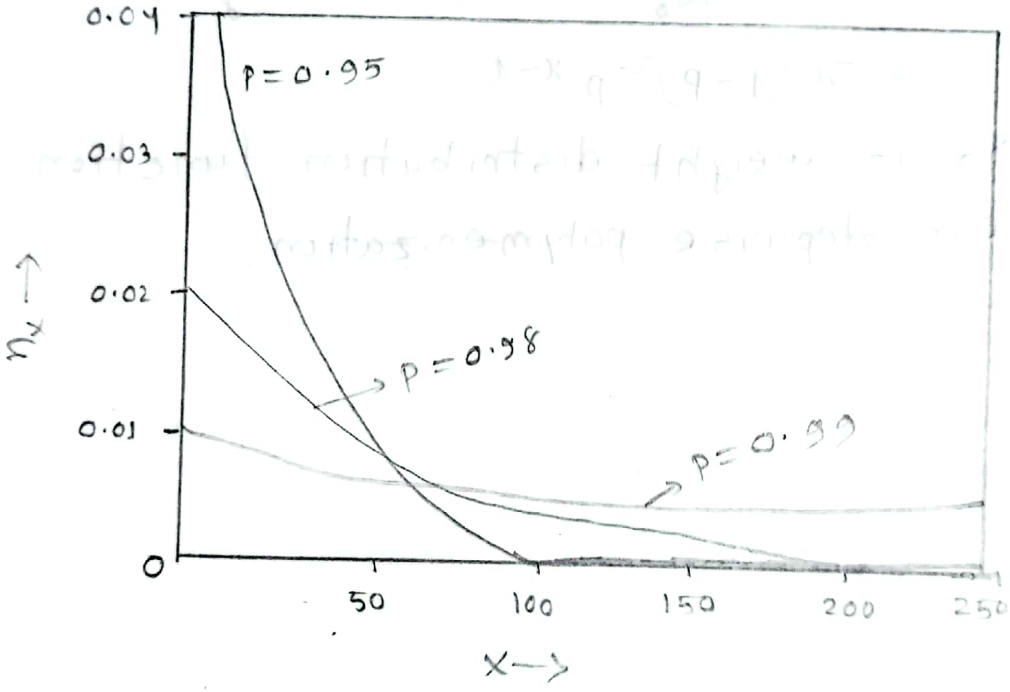


Fig.: Mole fraction distribution of Linear step-growth polymers.

* Weight distribution :

Again the weight distribution function w_x of x is given by -

$$w_x = \frac{x N_x}{N_0}$$

$$= \frac{x N_0 (1-p)^2 p^{x-1}}{N_0} \quad \text{using equation (11)}$$

$$= x(1-p)^2 p^{x-1}$$

This is weight distribution function for a linear stepwise polymerization.

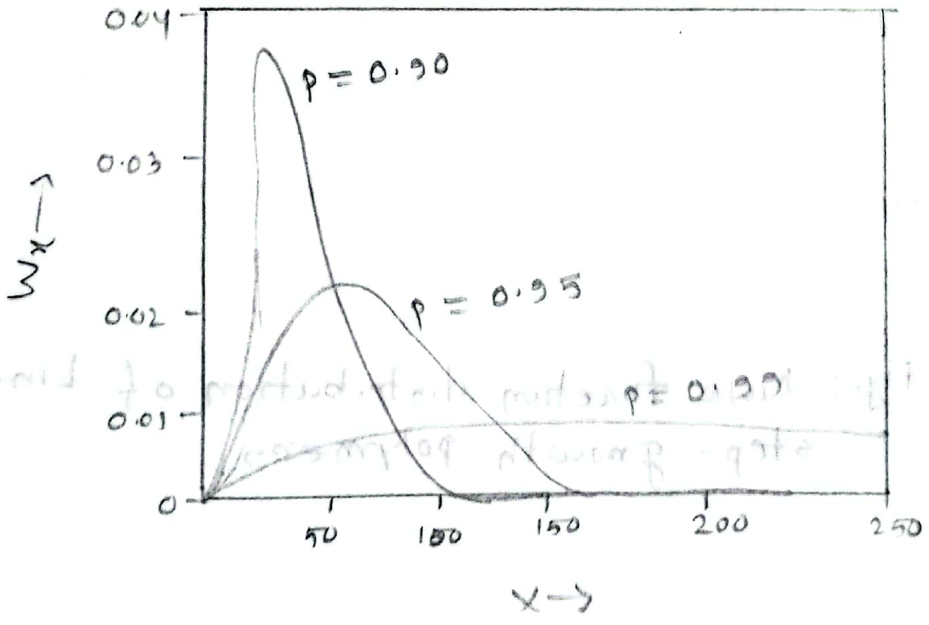


Fig.: Weight fraction distribution of Linear step growth polymerization.

Q. State the condition for high MW in step growth polymerization. [2017] [2016]

Molecular weight in step Growth polymerization is determined by the fractional conversion p of the monomer during polymerization.

The conditions necessary for higher molecular weight in step growth polymerization—

I. High conversions.
II. High monomer purities.
III. High reaction yields.
IV. Stoichiometric equivalance of functional group.

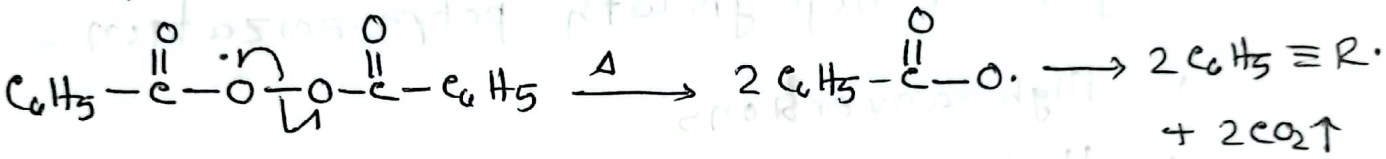
Q. How could you limit molecular weight in step Growth polymerization. [2016]

I. To stop the polymerization reaction by rapid cooling when the desired M.W is attained.
II. To use an excess amount of one monomer.
III. To use small amount of monofunctional reactant.

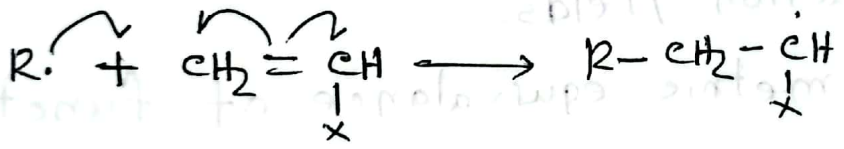
Q. Explain the initiation, propagation and termination steps with suitable examples for free radical polymerization. [2015] [2017]

Initiation:

Initiation of a chain reaction involves the formation of free radicals from an initiator. Benzoyl peroxide is commonly used initiator.

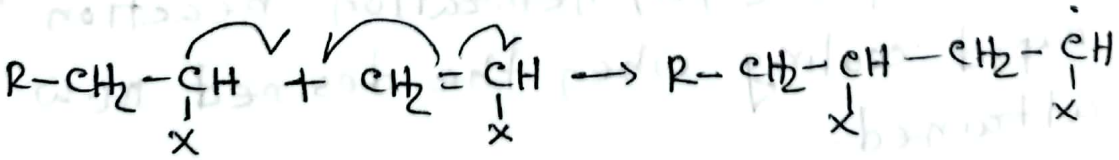


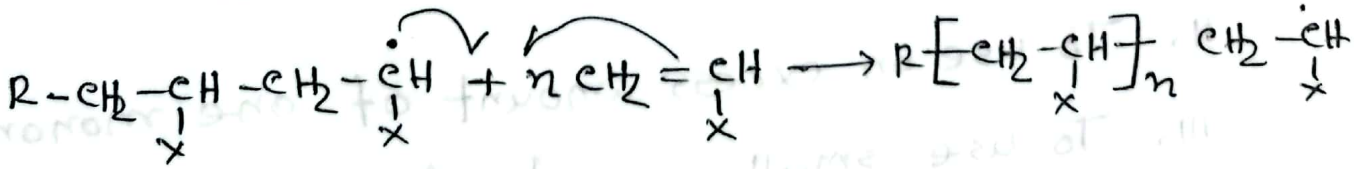
Or, peroxide → Radicals (R·)



propagation:

In propagation unit, the reaction site of the first monomer unit attacks the double bond of a fresh monomer.

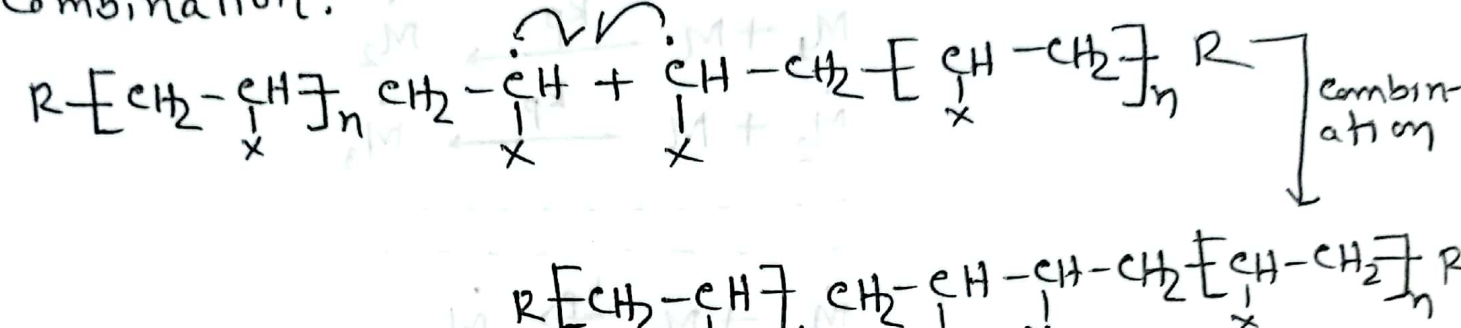




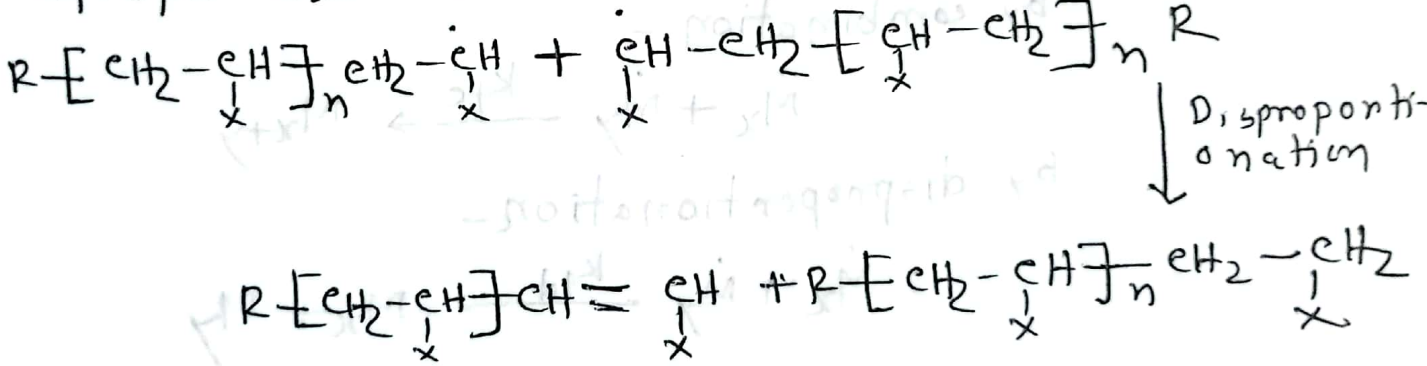
Termination:

Termination can take place either by the combination of two free radical to form one molecule or by transfer of a hydrogen atom in a disproportionation reaction.

1. Combination:



II. Disproportionation:



Q. Discuss the kinetics of free radical polymerization: [2015] (Lecture).

I. Initiation:

$$I \xrightarrow{k_d} 2\dot{R} \quad \text{---(I)}$$

$$\dot{R} + M \xrightarrow{k_a} \dot{M}_1 \quad \text{---(II)}$$

II. propagation:

$$\dot{M}_1 + M \xrightarrow{k_p} \dot{M}_2$$

$$\dot{M}_2 + M \xrightarrow{k_p} \dot{M}_3$$

$$\dots\dots\dots\dots$$

$$\dot{M}_x + M \xrightarrow{k_p} \dot{M}_{x+1}$$

III. Termination:

By combination –

$$\dot{M}_x + \dot{M}_y \xrightarrow{k_{tc}} M_{x+y}$$

By disproportionation –

$$\dot{M}_x + \dot{M}_y \xrightarrow{k_{td}} M_x + M_y$$

The rate of initiation –

$$v_i = \frac{d[\dot{M}]}{dt} = 2 f k_d [I]$$

Where, f represents the fraction of radicals formed by equation (I)

The rate of termination -

$$v_t = -\left(\frac{d[\dot{M}]}{dt}\right)_t$$

$$= 2k_t [\dot{M}]^2$$

According to steady state condition -

$$v_i = v_t$$

$$\Rightarrow 2f k_d [I] = 2k_t [\dot{M}]^2$$

$$\Rightarrow [\dot{M}]^2 = \frac{2f k_d [I]}{2k_t}$$

$$\Rightarrow [\dot{M}] = \left(\frac{f k_d [I]}{k_t}\right)^{1/2}$$

The rate of propagation,

$$v_p = -\frac{d[M]}{dt}$$

$$= k_p [M][\dot{M}]$$

$$= k_p \left(\frac{f k_d [I]}{k_t}\right)^{1/2} [M] \quad [\text{putting the value of } \dot{M}]$$

The rate of propagation, v_p is essentially the rate of polymerization and it is proportional to $[I]^{1/2}$ and $[M]$.

If the initiator efficiency is high f is independent of $[M]$ and if it is low then, $f \propto [M]$

Evaluation of rate constant,

$$\frac{k_p^2}{k_t} = \frac{2v_p^2}{v_i [M]^2}$$

Q. What is meant by inhibition and retandation? Mention one use of the inhibitor. [2015]

~~Inhibition~~: Retandation:

A retarder is a chemical substance that can react with a radical to form a stable product which is no capability of adding monomer and the process is called reterdation

Inhibition:

If the retarder is very effective, capable of inhibiting polymer chain growth and no polymer can form in their condition, then it is called inhibition.

Uses of inhibitors:

I. Reduce the concentration of radicals.

II. Shortend the average life of radical and thus length of polymer chain.

* Degree of polymerization and chain transfer:

kinetic chain length:

The kinetic chain length ν is defined as the number of monomer units consumed per active center. So,

$$\nu = \frac{v_p}{v_i} = \frac{v_p}{v_t}$$

$$\therefore \nu = \frac{k_p [M] [\dot{M}]}{2 k_t [\dot{M}]^2}$$

$$= \frac{k_p [M]}{2 k_t [\dot{M}]}$$

$$= \frac{k_p [M]}{2 k_t \left(\frac{f k_d [I]}{k_t}\right)^{1/2}}$$

$$= \frac{k_p [M]}{2 \left(f k_t k_d [I]\right)^{1/2}}$$

$v_p = k_p [M] [\dot{M}]$

$v_t = 2 k_t [\dot{M}]^2$

$\bar{x}_n = 2\nu$ for termination of combination

$\bar{x}_n = \nu$ for termination of disproportionation.

Chain transfer:

chain transfer reaction-

$$\dot{M}_x + X_p \longrightarrow M_x M + \dot{p}$$

$$\dot{p} + M \longrightarrow p\dot{M}$$

where, p may be monomer, initiator, solvent or other chain transfer agent.

Degree of polymerization.

$$\bar{x}_n = \frac{\text{rate of growth}}{\Sigma \text{rates of all reactions leading to dead polymer}}$$

$$= \frac{v_p}{f k_d [I] + k_{tr,M}[M][\dot{M}] + k_{tr,S}[S][\dot{M}] + k_{tr,I}[I][\dot{M}]}$$

Where the term represent termination by combination and transfer to monomer, solvent and initiator.

If termination by disproportion,

$$C_M = \frac{k_{tr,M}}{k_p}, \quad C_S = \frac{k_{tr,S}}{k_p}, \quad C_I = \frac{k_{tr,I}}{k_p}$$

Then,

$$\frac{1}{\bar{x}_n} = \frac{k_t}{k_p^2} \cdot \frac{v_p}{[M]^2} + C_M + C_S \frac{[S]}{[M]} + C_I \frac{k_t}{k_p^2 + k_d} \frac{v_p^2}{[M]^3}$$

In presence of solvent, the equation reduced to -

$$\frac{1}{\bar{X}_n} = \left(\frac{1}{\bar{X}_n}\right)_0 + C_s \frac{[S]}{M}$$

The linear dependence of $\frac{1}{\bar{X}_n}$ on $[S]/M$ is —

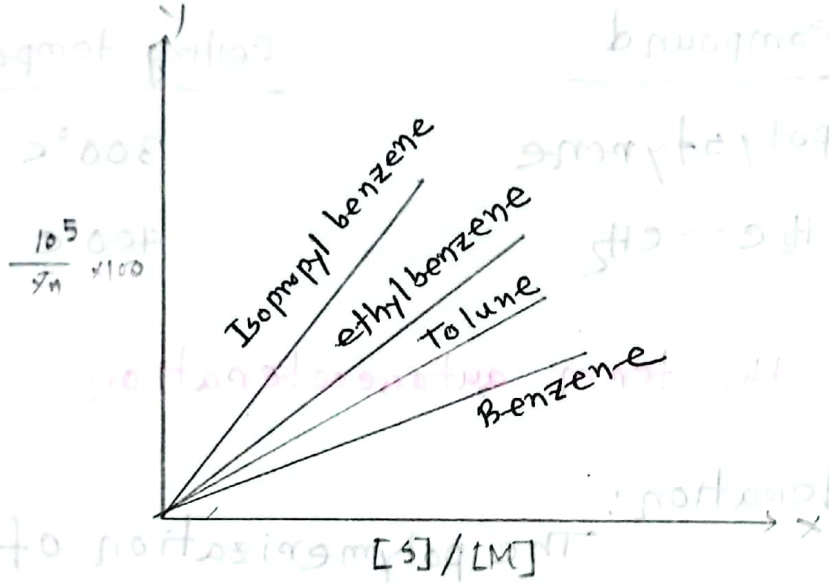


Fig.: Effect of chain transfer to solvent on the degree of polymerization.

* Ceiling temperature:

Ceiling temperature: The temperature at which rating of polymerization and depolymerization become equal is called ceiling temperature.

Compound	Ceiling temperature
polystyrene	300°c
$H_2C=CH_2$	400°c.

Q. Define the term autoacceleration.

Autoacceleration: The polymerization of certain monomers in undiluted or in concentrated solution is deviated from first-order kinetics in the direction of an increase in reaction rate and molecular weight is termed autoacceleration or the gel effect.

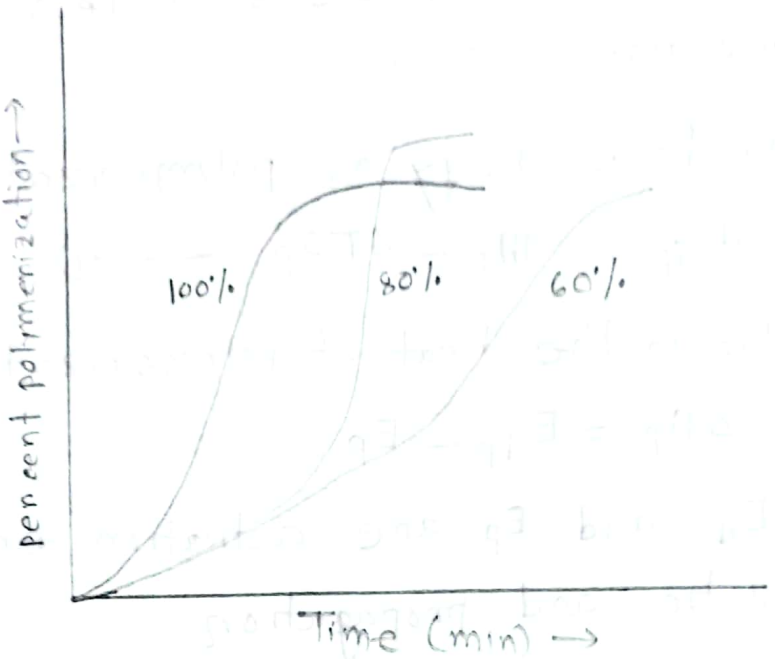


Fig.: Autoacceleration of polymerization.

Q. Discuss the thermodynamics of free radical polymerization. [2017]

Gibbs free energy of polymerization,

$$\Delta G_p = \Delta H_p - \Delta T S_p \quad \text{——(1)}$$

Where ΔH_p is the heat of polymerization.

$$\Delta H_p = E_{dp} - E_p$$

Where, E_{dp} and E_p are activation energy of depolimerization and propagation.

ΔG_p is negative at low temperature.
and high temperature, it is less negative.

But at ceilling temperature (T_c), $\Delta G_p = 0$

From equation (1) we get,

$$T_c = \Delta H_p / \Delta S_p.$$

Q. Explain the difference between step-reaction polymerization and chain reaction polymerization. [2016].

Step reaction polymerization	chain reaction polymerization
i. Step polymerization is a process of polymerization between polyfunctional monomers by various condensation and addition reactions.	i. Chain polymerization is a polymerization in which proceeds exclusively by reaction between monomer and active site on the polymer chain with regeneration of active site at the end of each growth step.
ii. Any two molecular species can react.	ii. Growth occurs only by addition of monomer to active chain end.
iii. Monomers disappears early in the reaction.	iii. Monomer concentration decreases steadily thoughout reaction.
iv. Similar steps repeated thoughout reaction process.	iv. Different steps operate at different stages of mechanism.
v. Degree of polymerization low to moderate.	v. Degree of polymerization can be very high.
vi. Average M.w for the reaction increases throughout the polymerization.	vi. Average chain length remains approximately constant throughout the polymerization

vii. High extent of reaction are required to obtain high chain lengths.	vii. As reaction time is increased, polymer yield increases but M.w remains same.
viii. No initiator necessary	viii. Initiator required.
ix. Only monomers and polymers are observed.	ix. Any type of molecules can be observed